DIMENSION FORMULA ALONG SMOOTH MORPHISMS OF ALGEBRAIC STACKS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

Source reference. Original lemma 0DRI: sources/stacks-chow/tags.txt, source lines 17223–17223. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0DRN: sources/stacks-chow/tags.txt, source lines 17228–17228. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *If $f : \mathcal{U} \to \mathcal{X}$ is a smooth morphism of locally Noetherian algebraic stacks, and if $u \in |\mathcal{U}|$ with image $x \in |\mathcal{X}|$, then*

$$\dim_u(\mathcal{U}) = \dim_x(\mathcal{X}) + \dim_u(\mathcal{U}_x).$$

Proof. Choose a smooth surjective morphism $V \to \mathcal{U}$ whose source is a scheme, and let $v \in |V|$ be a point mapping to u. Then the composite $V \to \mathcal{U} \to \mathcal{X}$ is also smooth, and by Lemma 0DRI we have $\dim_x(\mathcal{X}) = \dim_v(V) - \dim_v(V_x)$, while $\dim_u(\mathcal{U}) = \dim_v(V) - \dim_v(V_u)$. Thus

$$\dim_u(\mathcal{U}) - \dim_x(\mathcal{X}) = \dim_v(V_x) - \dim_v(V_u).$$

Choose a representative $\operatorname{Spec} k \to \mathcal{X}$ of x and choose a point $v' \in |V \times_{\mathcal{X}} \operatorname{Spec} k|$ lying over v, with image u' in $|\mathcal{U} \times_{\mathcal{X}} \operatorname{Spec} k|$; then by definition $\dim_u(\mathcal{U}_x) = \dim_{u'}(\mathcal{U} \times_{\mathcal{X}} \operatorname{Spec} k)$, and $\dim_v(V_x) = \dim_{v'}(V \times_{\mathcal{X}} \operatorname{Spec} k)$.

Now $V \times_{\mathcal{X}} \operatorname{Spec} k \to \mathcal{U} \times_{\mathcal{X}} \operatorname{Spec} k$ is a smooth surjective morphism (being the base-change of such a morphism) whose source is an algebraic space (since V and $\operatorname{Spec} k$ are schemes, and $\mathcal{X}$ is an algebraic stack). Thus, again by definition, we have

$$\begin{aligned} \dim_{u'}(\mathcal{U} \times_{\mathcal{X}} \operatorname{Spec} k) &= \dim_{v'}(V \times_{\mathcal{X}} \operatorname{Spec} k) - \dim_{v'}(V \times_{\mathcal{X}} \operatorname{Spec} k)_{u'}) \\ &= \dim_v(V_x) - \dim_{v'}((V \times_{\mathcal{X}} \operatorname{Spec} k)_{u'}). \end{aligned}$$

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Now $V \times_{\mathcal{X}} \operatorname{Spec} k \cong V \times_{\mathcal{U}} (\mathcal{U} \times_{\mathcal{X}} \operatorname{Spec} k)$, and so Lemma 0DRN shows that $\dim_{v'}((V \times_{\mathcal{X}} \operatorname{Spec} k)_{u'}) = \dim_v(V_u)$. Putting everything together, we find that

$$\dim_u(\mathcal{U}) - \dim_x(\mathcal{X}) = \dim_u(\mathcal{U}_x),$$

as required. $\square$